

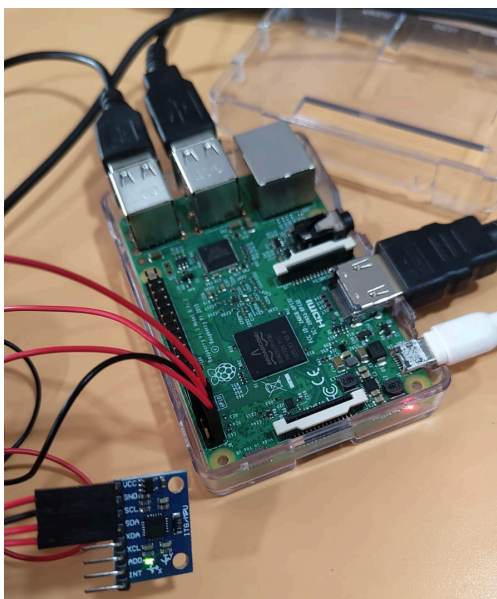
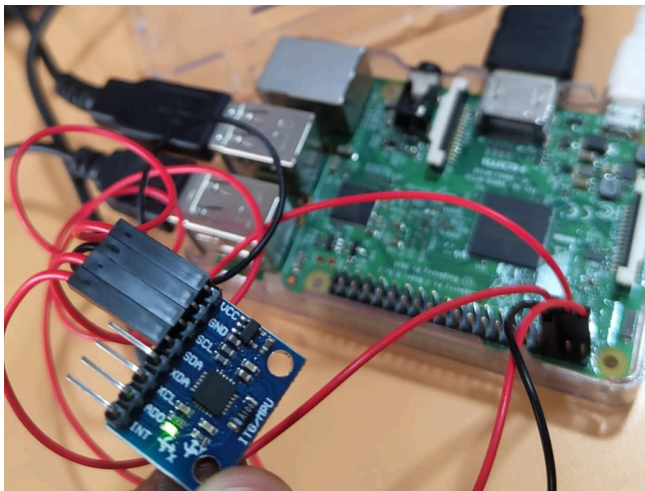
Practical No.7

Aim:- Analog and Digital Sensors

1. Connect sensor MPU-6050 Module 3 Axis Gyroscope Accelerometer Module and RaspberryPi.

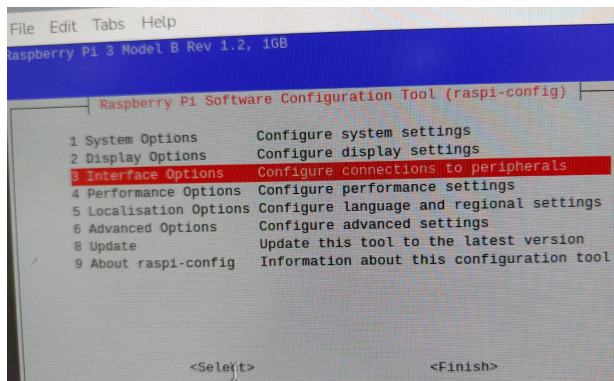
The MPU-6050 module typically uses four pins for a basic connection to the Raspberry Pi's GPIO header.

- **VCC:** Connect to Raspberry Pi **3.3V (Pin 1)**.
- **GND:** Connect to Raspberry Pi **Ground (Pin 6 or 9)**.
- **SDA:** Connect to Raspberry Pi **SDA (Pin 3, GPIO 2)**.
- **SCL:** Connect to Raspberry Pi **SCL (Pin 5, GPIO 3)**.

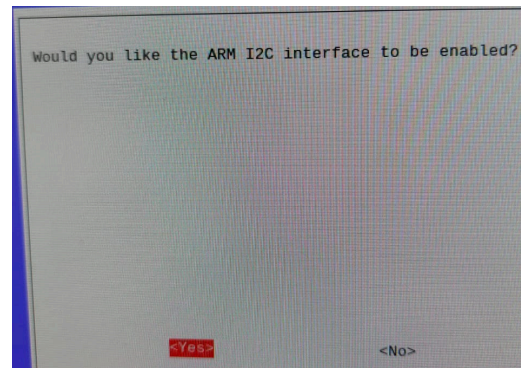
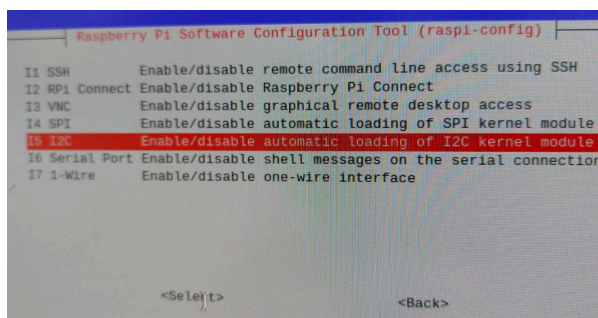


2. Enable I2C in Raspberry Pi using:

`sudo raspi-config`



Then enable I2C from Interface Option



3. Create Folder:-

`mkdir mpu_project`

`cd mpu_project`

Create Python Environment:-

`python3 -m venv myenv.`

`source myenv/bin/activate`

Install libraries: `mpu6050`, `requests`.

`pip install mpu6050-raspberrypi requests`

4. Write python code and save it in the `mpu_project` directory.

```
motor1.py axiss.py
1 from mpu6050 import mpu6050
2 import time
3
4 mpu = mpu6050(0x68)
5
6 while True:
7     print("TIME :" + str(mpu.get_temp()))
8     print()
9
10    accel_data = mpu.get_accel_data()
11    print("acc x :" + str(accel_data['x']))
12    print("acc y :" + str(accel_data['y']))
13    print("acc z :" + str(accel_data['z']))
14    print()
15
16    gyro_data = mpu.get_gyro_data()
17    print("gyro x :" + str(gyro_data['x']))
18    print("gyro y :" + str(gyro_data['y']))
19    print("gyro z :" + str(gyro_data['z']))
20    print()
21
22    print("-----")
23    time.sleep(1)
24    |
```

Output:-

When the sensor is rotated, acceleration and gyroscope readings are displayed continuously on the terminal. The values change according to the movement and orientation of the sensor.

 |     raspberrypi@raspber..

File Edit Tabs Help

```
acc x :2.3247698120117186
acc y :3.2249896362304686
acc z :5.944802722167968
```

```
gyro x :122.22137404580153
gyro y :-249.93893129770993
gyro z :194.05343511450383
```

TIME :47.777058823529416

```
acc x :6.610390783691406
acc y :10.94868419189453
acc z :16.65646094970703
```

```
gyro x :-18.14503816793893
gyro y :97.2442748091603
gyro z :-59.69465648854962
```

TIME :47.777058823529416

```
acc x :1.4796166259765624
acc y :-3.380612744140625
acc z :14.389151977539061
```

```
gyro x :-43.83969465648855
gyro y :250.12977099236642
gyro z :77.56488549618321
```

TIME :47.63588235294118

```
acc x :4.714183068847656
acc y :4.7381250854492185
acc z :7.802703210449218
```

```
gyro x :24.6793893129771
gyro y :47.44274809160305
gyro z :14.816793893129772
```

TIME :47.63588235294118

```
acc x :4.113238452148437
acc y :4.922479612304105
```